

BioMetAll: Identifying Metal-Binding Sites in Proteins from Backbone Preorganization

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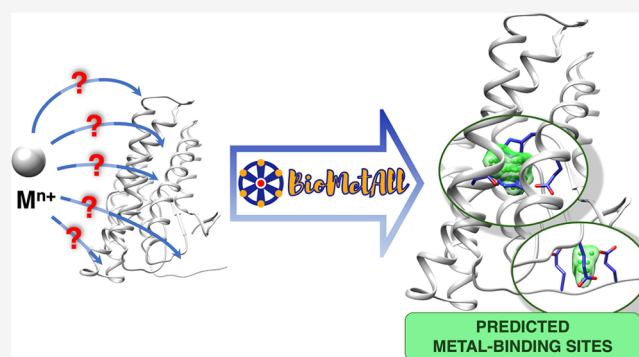
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ABSTRACT: With a large amount of research dedicated to decoding how metallic species bind to proteins, *in silico* methods are interesting allies for experimental procedures. To date, computational predictors mostly work by identifying the best possible sequence or structural match of the target protein with metal-binding templates. These approaches are fundamentally focused on the first coordination sphere of the metal. Here, we present the BioMetAll predictor that is based on a different postulate: the formation of a potential metal-binding site is related to the geometric organization of the protein backbone. We first report the set of convenient geometric descriptors of the backbone needed for the algorithm and their parameterization from a statistical analysis. Then, the successful benchmark of BioMetAll

on a set of more than 90 metal-binding X-ray structures is presented. Because BioMetAll allows structural predictions regardless of the exact geometry of the side chains, it appears extremely valuable for systems whose structures (either experimental or theoretical) are not optimal for metal-binding sites. We report here its application on three different challenging cases: (i) the modulation of metal-binding sites during conformational transition in human serum albumin, (ii) the identification of possible routes of metal migration in hemocyanins, and (iii) the prediction of mutations to generate convenient metal-binding sites for *de novo* biocatalysts. This study shows that BioMetAll offers a versatile platform for numerous fields of research at the interface between inorganic chemistry and biology and allows to highlight the role of the preorganization of the protein backbone as a marker for metal binding. BioMetAll is an open-source application available at <https://github.com/insilichem/biometall>.



INTRODUCTION

Metal ions, including transition ones, play a fundamental role in living organisms. Present in about 30% of the known genome,¹ their impacts on molecular and cellular processes are countless. Because biometallic complementarities sustain many mechanisms in living systems, they are essential for their adaptation to the environment and key to their health. Understanding how metallic moieties interact with biological systems is not just a health-related question. It is also a source of inspiration for the development of new synthetic routes, including environmentally friendly ones. As such, chemists have long intended to mimic biometallic synergies for industrial purposes. In the form of small amenable chemical systems (biomimetics) or complex *de novo* enzymes, a substantial part of chemical efforts focuses on reproducing or even extending nature's bioinorganic repertoire.

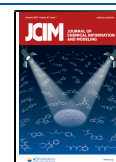
The specific problem of predicting where and how metallic moieties bind to proteins is of crucial importance in numerous areas of chemistry and its closest interfaces.² After decades, researchers in bioinorganics still intensively work on decoding the exact weight of the molecular variables that are involved in

the construction of biometallic edifices (charges of the metal ions and their amino acid environment vs electric field of the entire protein, redox properties of the metal, etc.). *In silico* approaches offer an excellent option to get a better insight into metal–protein interactions and help in establishing predictive patterns. To date, however, only a limited ensemble of computational tools is available. These methods are either based on sequence or structural analyses.^{3–5} Both families of methods could be used conjointly.

Computational predictors of metal-binding sites built on sequence analyses are mostly based on scanning the sequence of a target protein to identify those regions where amino acid patterns match a metal-binding site fingerprint. Some examples are MetalDetector,⁶ which identifies potential transition-metal-

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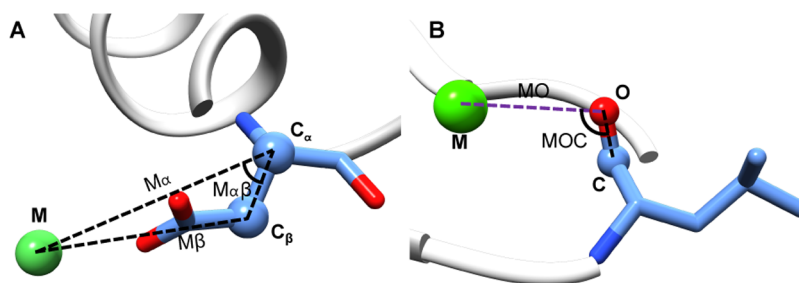


Figure 1. Geometric features analyzed for all the structures of MetalPDB. (A) Geometric features considered for coordination bonds with a side-chain donor. The metal (M) is depicted in a green ball, and α -carbon (C_α) and β -carbon (C_β) are depicted in blue balls. The distance from the metal to the α -carbon ($M\alpha$) and the distance from the metal to the β -carbon ($M\beta$) are represented in black dashed lines. The angle between the metal, α -carbon, and β -carbon is labeled as $M\alpha\beta$. (B) Geometric features considered for coordination bonds with a backbone oxygen donor. The metal (M) is depicted in a green ball, the backbone oxygen atom (O) is depicted as a red ball, and the backbone carbon (C) is depicted in a blue ball. The distance from the metal to the backbone oxygen (MO) is represented in a purple dashed line. The angle between the metal, backbone oxygen, and backbone carbon is labeled as MOC.

binding sites involving Cys and His side chains as donors; MetalPredator,⁷ which predicts potential Fe–S motifs in proteins; and ZincFinder,⁸ which aims at predicting zinc-binding motifs. Structure-based predictors, on their side, focus on identifying specific structural metal-binding motifs derived from databases of existing proteins bound to metal structures that match the best with the atomic organization of the target protein. Some of these approaches are metal-specific, such as TEMPS and FUTURE, which are dedicated to identifying zinc-binding motifs.^{9,10} Others consider more extensive lists of metals that typically include Ca, Cu, Fe, Mg, Mn, and Zn.^{11–17}

Metal-binding site predictors are successful in finding sites that match well-known patterns. However, the identification of those sites in systems that do not display an optimal sequence or whose structure does not present the sufficient amount of preorganization can be challenging. Such situations are frequent in modern inorganic biochemistry. For example, *de novo* design of metalloenzymes frequently requires mutations of the protein to generate a convenient coordination center. Moreover, experimental structures may only provide with metal-free (*apo*) forms without convenient organization of the protein side chain for the metal to bind, including transient geometries that require conformational changes or modification of the coordination environment (vacant sites, presence of labile ligands, etc.).

In recent years, our group has actively worked in areas where it was essential to predict with confidence those amino acids that could coordinate metal compounds either for proteins¹⁸ or peptides.¹⁹ In these works, we frequently confront with new-to-nature metal-binding sites, X-ray structures of proteins without a well-defined binding site,^{20,21} or even putative binding sites of metallodrugs.^{22–29} A part of our efforts allowed us to develop a series of docking approaches^{30–32} that can well reproduce low-energy poses of metal–protein complexes. However, such approaches are only viable if good guesses for metal-binding regions are accessible on the first hand. For this aspect, in a first statistical analysis about 10 years ago,²⁰ we observed that (i) any amino acid with its α -carbon located within a sphere up to 7–9 Å could potentially be considered as a metal ligand (obviously considering possible coordination by residues with long side chains such as lysine or tyrosine) and (ii) the center of mass of the coordinating residues generally does not stand far from the exact position of the metal ions observed in the X-ray structure. In a subsequent study,³² we preselected docking areas by filtering the distance from the

metal to the β -carbon of the amino acids Asp, His, Glu, and Cys using the range from 2.5 to 5.0 Å. Both works showed that few geometric descriptors related to the backbone of the protein afford convenient information to predict metal-binding sites rather than focusing only on the first coordination sphere donors (mainly side chains) that coordinate the metal.

Here, we present BioMetAll, a comprehensive piece of software that is based on our observations that backbone geometries could be used as a preorganization marker for metal-binding sites and so could be used as a predictive tool. To the difference of our previous works where the preorganization hypothesis was partially developed (mostly by straightforward analyses for a limited number of amino acids and based on very general metrics), BioMetAll accounts for extensive applications of the preorganization concept. This is concretized as an entirely self-contained program that is readily accessible to the community based on an analysis on more than 170,000 biological metal-binding complexes, the consideration of all possible amino acids, and a more comprehensive set of geometric variables, such as distances and angles involving the metal and α - and β -carbons. It represents a major step forward in using the preorganization hypothesis and how to predict metal-binding sites in proteins.

The program accurately reproduces the experimental metal-binding motifs on a benchmark of 93 structures. Furthermore, we present three specific case studies to illustrate its potential. BioMetAll is efficient to (i) predict metal-binding sites with particular amino acid motifs even for noncanonical metal-coordinating amino acids and backbone atoms; (ii) determine transient metal-binding sites in structures that need conformational changes to reach the complete coordination sphere, which could be related to metal diffusion pathways; and even (iii) predict potential mutations of an existing protein to generate convenient metal-binding sites. This new approach can make a significant contribution to future development in fields such as metalloenzyme and metallodrug design.

■ COMPUTATIONAL METHODOLOGY

Concept. BioMetAll relies on the assumption that the geometric patterns of the backbone of a given protein environment encode sufficient information on its preorganization to coordinate metal ions. By doing so, predictions are less conditioned by finding perfect matches and so could detect those binding sites that are partially organized (e.g., missing ones or several well-positioned side chains) or even help in

Table 1. Ranges (Excluding Outliers) for Each One of the Geometric Features Analyzed, Grouped by the Type of Amino Acid^a

amino acid	percentage of entries ^b (%)	distance $M\alpha$ (Å)	distance $M\beta$ (Å)	angle $M\alpha\beta$ (rad)
Asp	28.63	[3.907–6.192]	[3.658–5.052]	[0.003–1.871]
His ^c	23.36	[3.076–7.098]	[3.047–6.073]	[0.001–2.018]
Glu	18.96	[3.591–8.303]	[4.123–6.108]	[0.000–2.305]
Cys	18.01	[3.248–5.451]	[2.794–3.829]	[0.006–1.734]
Asn	3.80	[4.048–5.701]	[3.866–5.250]	[0.682–1.697]
Thr	1.99	[3.313–5.096]	[2.530–4.371]	[0.184–1.647]
Ser	1.81	[3.384–5.171]	[2.587–4.353]	[0.163–1.521]
Gln	1.26	[3.217–7.835]	[3.512–6.565]	[0.012–2.268]
Met	0.72	[4.426–6.706]	[3.654–5.094]	[0.016–1.228]
Tyr	0.71	[6.896–9.016]	[6.367–8.087]	[0.501–1.433]
Lys	0.34	[5.134–9.280]	[4.358–7.997]	[0.029–1.918]
Arg	0.30	[3.409–10.162]	[2.913–9.042]	[0.024–1.991]
Leu	0.03	[3.784–4.678]	[4.353–5.622]	[1.464–2.470]
Trp	0.02	[3.912–7.478]	[4.244–6.573]	[0.471–2.703]
Ile	0.02	[4.236–4.817]	[5.039–5.702]	[1.826–2.169]
Val	0.01	[3.517–5.003]	[4.358–5.722]	[1.639–2.166]
Ala	0.01	[3.402–5.126]	[4.254–5.991]	[1.584–2.368]
Phe	0.00	[3.499–4.922]	[4.289–5.590]	[1.666–2.300]
Pro	0.00	[3.646–4.529]	[4.860–5.246]	[1.746–2.678]

^aCoordinations with side-chain donors were considered in this analysis. Outliers were excluded from obtaining each range. With IQR being the interquartile range and Q_x being the quartile x , a point is considered an outlier if it is below ($Q_1 - 1.5 \times \text{IQR}$) or above ($Q_3 + 1.5 \times \text{IQR}$).

^bPercentage of bonds corresponding to each residue of the total of 400,077 entries analyzed. ^cRanges for His were obtained from merging the subranges calculated for His_δ and His_ε tautomers.

designing new metal coordination environments throughout mutations (e.g., for applications in enzyme design).

BioMetAll prediction works by embedding a protein in a grid of fictitious metal probes, which are further evaluated using geometric features obtained for each amino acid by statistical analysis. The entire set of geometric parameters is divided into two: one for coordinating atoms being a part of the side chains and the other for atoms being a part of the backbone. For side chains, the geometric features are as follows (Figure 1A): $M\alpha$, the distance from the metal to the α -carbon; $M\beta$, the distance from the metal to the β -carbon; and $M\alpha\beta$, the angle between the metal, the α -carbon, and the β -carbon. For backbone oxygen atoms, the geometric features are as follows (Figure 1B): MO, the distance from the metal to the backbone oxygen, and MOC, the angle between the metal, the backbone oxygen, and the backbone carbon.

Statistical Analysis. We analyzed the entire bunch of structures available in MetalPDB, a publicly accessible database of preselected metal-containing proteins from the Protein Data Bank (PDB).^{33,34} This analysis was carried out thanks to an in-house PyChimera^{35,36} script applied on ca. 171,000 metal-binding sites. It ended up with a list of ca. 500,000 entries (Table S1) after cleaning the database from metal-donor distances lower than 1.5 Å or greater than 3.0 Å.³⁷

From the 500,000 entries, a range of values were obtained for each geometric feature and amino acid (Table 1, Figure 2A–C). For histidine, the subranges obtained for His_δ and His_ε tautomers were merged. From the analysis grouping by metal (Figure 2D–F), it became apparent that the ranges exhibited a considerable overlap, justifying the use of general ranges for metals. Although the analysis of coordination by backbone atoms was undertaken for each amino acid, the final results (Table S2) presented a large overlap. In consequence, a global range for MO and MOC was considered for all the entries: MO, the distance metal–backbone oxygen from 1.809

to 3.000 Å and MOC, the angle metal–backbone oxygen–backbone carbon from 1.774 to 3.139 rad.

Further analyses were conducted to provide a series of default parameters for a prototypical prediction. It first emerges that 80% of donor atoms are a part of the side chains (Table S3), and 19% are a part of the backbone (Table S4). Default calculations in BioMetAll were therefore set to look for side chains primarily and left backbone coordination as optional. For side chains, the 89% of coordinating atoms come from what we called *canonical* metal-coordinating amino acids, namely, Asp, His, Glu, and Cys, while a 10.9% belongs to *noncanonical* amino acids (Asn, Thr, Ser, Gln, Met, Tyr, Lys, and Arg). Only the last small fraction (<0.1%) are coordinations with a carboxylate terminal amino acid (Table S3). Finally, most common metal-binding sites account for three coordinating amino acids on average (with an interquartile range of 2). On these grounds, BioMetAll works with default parameters that involved *canonical amino acid* side chains with a number of amino acids *per metal-binding sites of three*. The user can configure any other possible situation in a user-friendly manner.

It is to notice that besides standard amino acids, exogenous ligands (e.g., water or small molecules) can participate in the coordination of the metal. Although metrics for BioMetAll parameters have been obtained purely on amino acid geometric magnitudes, exogenous ligands could participate in the coordination of the metal on the analyzed set. The calculations we performed so far show that BioMetAll implicitly takes into account the changes in the metal position that these exogenous ligands could cause (as shown later in the article).

Overview of BioMetAll Workflow. The BioMetAll workflow (Figure 3) is divided into three sequential steps.

Generation of a Probe Grid. An input .pdb file (that can be downloaded from the same server of the PDB³⁸) is parsed to retain only the Cartesian coordinates of α -carbons, β -carbons,

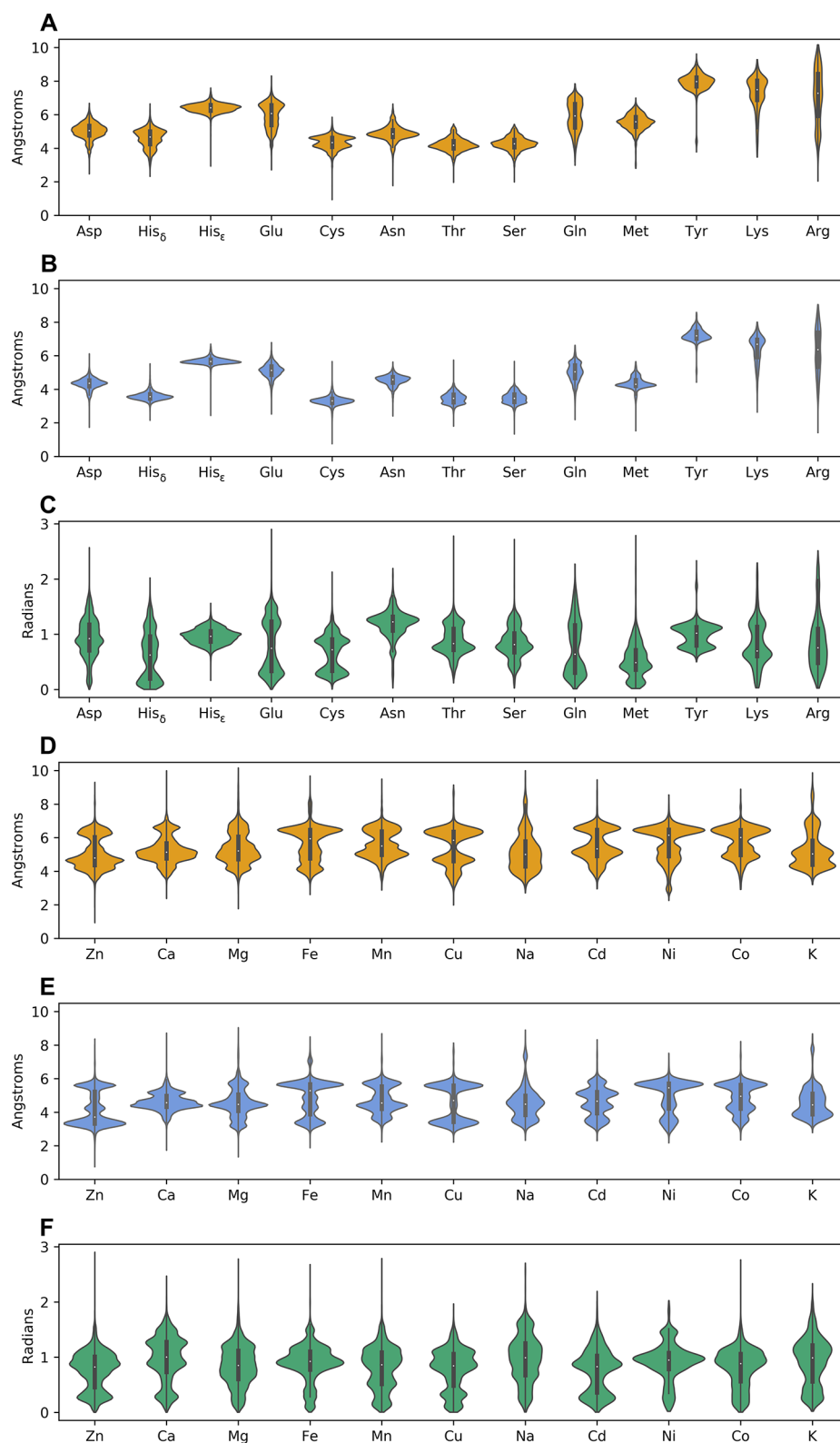


Figure 2. Violin plots for the different features studied in the statistical analysis. Coordination bonds between the metal and a side-chain atom are considered in all cases. (A) Distance from the metal to α -carbon of the coordinating amino acid for *canonical* and *noncanonical* amino acids. (B) Distance from the metal to β -carbon of the coordinating amino acid for *canonical* and *noncanonical* amino acids. (C) Angle between the metal, α -carbon, and β -carbon of the coordinating amino acid for *canonical* and *noncanonical* amino acids. (D) Distance from the metal to α -carbon of the coordinating amino acid for the 11 most representative metals. (E) Distance from the metal to β -carbon of the coordinating amino acid for the 11 most representative metals. (F) Angle between the metal, α -carbon, and β -carbon of the coordinating amino acid for the 11 most representative metals.

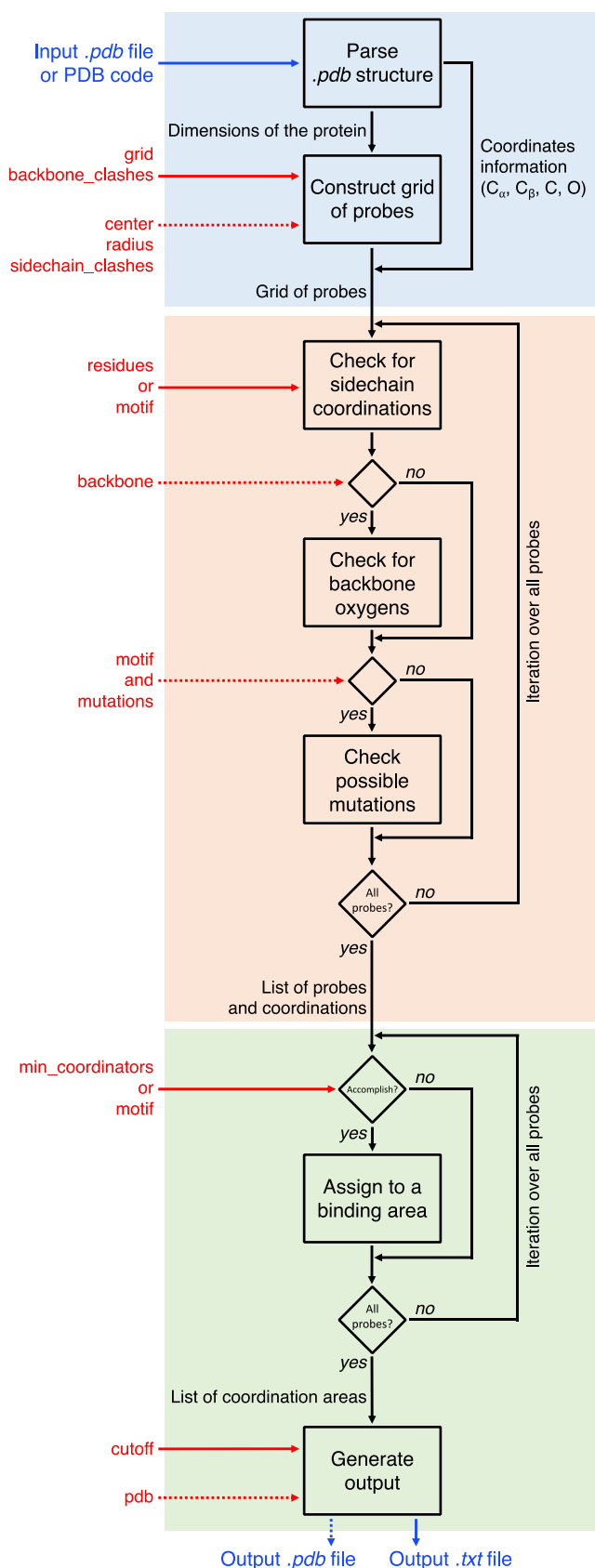


Figure 3. BioMetAll workflow. User-configurable parameters are colored in red (optional parameters have a dashed arrow). Inputs and outputs are colored in blue. The three main steps of the algorithm are highlighted in blue, orange, and green backgrounds.

backbone oxygens, and backbone carbons. Subsequently, a spherical grid of equidistributed points (i.e., probes) is generated, which by default embeds the entire protein with an extra margin of 8 Å (to account for superficial amino acids). The user can customize this sphere if interested in a portion of the system only. Finally, to avoid areas already occupied by the backbone of the protein, the program discards by default probes at less than 1.0 Å from any atom in the backbone. The user can configure this distance and also set an additional filter for the closeness to the side-chain atoms (see Table 2).

Table 2. User-Configurable Parameters for a BioMetAll Calculation

parameter	default value	utility
grid	1.0	distance (in Å) between probes in the grid
center	none	optional. Coordinates of the sphere center, when the user wants a custom search zone
radius	none	optional. Radius (in Å) of the search sphere, when the user wants a custom search zone.
residues	[Asp, His, Glu, Cys]	list of residues that will be considered as potential side-chain donors
backbone	none	list of residues that will be considered as potential backbone donors
motif	none	optional. Concrete structural motif to search
mutations	none	optional. Propose mutations to complete a motif
min_coordinators	3	minimum number of amino acids that a coordination environment must have to be valid
backbone_clashes	1.0	discards probes nearer than the given distance from any backbone atom
sidechain_clashes	0.0	optional. Discards probes nearer than the given distance from any side-chain atom
cutoff	0	cuts the solutions proposed by the software to a given threshold of the number of probes
pdb	no	generates a .pdb file with the proposed binding areas ready to open with a visualization program

Checking Geometric Criteria and User-Defined Constraints. The second stage of the process assigns a list of potentially coordinating amino acids to each probe. For every probe, the program checks which amino acids meet all the geometric ranges established in our statistical analysis, discarding those that fail in any restriction. Also, the user-defined restraints are applied at this stage, which allow for limiting the search to a specific list of amino acids, consider or not coordinations with the backbone oxygens, search for a particular structural motif, or even output feasible mutations which will be suitable to complete that motif (a list of parameters is provided in Table 2). This process concludes with a list of valid probes along with their potentially coordinating amino acids (see the example in Table 3).

Generation of Binding Areas and the Output. Once the potentially coordinating amino acids are assigned to each probe, they are considered the “coordinating environment” for that probe. The probes that share the same list of potentially coordinating amino acids are grouped. This provides a list of valid environments, each associated with a series of probes, constituting a zone in space (i.e., binding site) where a metal could bind the amino acids of the environment (see the example in Table 4). The result can be reported into a .pdb format that can further be visualized with any visualization program. The resulting coordination area/areas is/are ordered

Table 3. Examples of Valid Probes for a Search Restrained to a Minimum of Three Coordinating Amino Acids among HIS, ASP, CYS, and GLU

coordinates [x, y, z] of the probe	valid amino acids ^a	result of the check
[0.0, 0.0, 0.0]	[HIS23, ASP25, CYS78]	valid probe
[0.0, 0.0, 5.0]	[HIS23, ASP25]	discarded (only two aa)
[0.0, 5.0, 0.0]	[HIS23, ASP25, CYS78]	valid probe
[0.0, 5.0, 5.0]	[HIS23, ASP25, MET82]	discarded (MET is not in the user-defined list)
[5.0, 0.0, 0.0]	[CYS78, HIS80, GLU92]	valid probe
[5.0, 0.0, 5.0]	[CYS78, HIS80, GLU92]	valid probe
[5.0, 5.0, 0.0]	[CYS78, HIS80, HIS94]	valid probe
[5.0, 5.0, 5.0]	none	discarded

^aList of amino acids that fulfill all the geometric criteria.**Table 4.** Examples of Valid Coordinating Environments from the Probes Selected in the Examples of Table 3, with the List of Probes (i.e., Binding Sites) Associated with Each Environment

coordinating environment	list of probes (binding sites)
[HIS23, ASP25, CYS78]	[0.0, 0.0, 0.0], [0.0, 5.0, 0.0]
[CYS78, HIS80, GLU92]	[5.0, 0.0, 0.0], [5.0, 0.0, 5.0]
[CYS78, HIS80, HIS94]	[5.0, 5.0, 0.0]

by decreasing number of probes and saved in an additional output text file. It is important to bear in mind that BioMetAll does not return a unique spot but groups of probes that constitute the areas where a metal could bind.

Availability of the Code. BioMetAll is an open-source command-line application implemented in Python 3.7 language with NumPy³⁹ and psutil being the only required dependencies. The user manual and tutorials are available at <https://github.com/insilichem/biometall>. It can be run as usual with any Python script once the dependencies mentioned above are installed on the system. Prebuilt executables (generated with PyInstaller⁴⁰) for Linux, Mac OS, and Windows 10 are available for direct use of BioMetAll without the need for any installation process.

RESULTS AND DISCUSSION

Motif Search Benchmark. The predictiveness of BioMetAll was tested for the detection of the two-histidine one-carboxylate motifs (FTMs) in a set of 53 crystallographic structures from the PDB. The FTM was chosen because it is a coordination sphere motif present in the active site of many metalloenzymes, such as carboxypeptidase A or naphthalene-dioxygenase.^{41–43} Thus, it is interesting to benchmark BioMetAll predictions against a set of experimental structures containing this motif.⁴⁴ This set appeared very convenient because it shows structures with a wide range in size (from 108 to 1074 residues) and great diversity of metals (Cd, Co, Cu, Fe, Hg, Mg, Mn, Ni, Ru, and Zn).

On average, BioMetAll returned 10 predicted coordinating environments for each crystallographic structure. The solutions are ordered by decreasing number of probes, with 25 ± 10 probes being the average of the solutions that exactly match the crystallographic motif. Because the probes are equidistributed points in space, it implies that the best-ranked solutions are those with the largest volume. We therefore consider that it is interesting to evaluate how reliable this sorting is for the prediction of an X-ray structure.

As a first metric to evaluate the correctness of the solutions, the ubication of the crystallographic binding site is considered. In this case, a solution would be correct if it shares at least one amino acid with the experimental motif. In 100% of the structures, there is a solution located at the binding site, the most populated solution being valid in 75% of the cases, and among the top five in 98% of the cases. Obviously, the other metric to consider is the prediction of the exact crystallographic motif implying both the location and the entire set of amino acids that coordinate the metal. In this case, the 100% of the FTMs were found among the solutions provided by BioMetAll, being the first solution in 40% of the cases, and among the top five in 93% of the structures analyzed. Detailed benchmark results are provided in Table S5. These ranges are not far from those of typical standard protein–ligand docking procedures.

The accuracy of the predicted binding area was further assessed by measuring the distance from the experimental metal position to the nearest computed probe. On average, the difference between calculated and experimental positions of the metal is about 0.56 ± 0.19 Å. Importantly, such quality is not influenced by the presence of donors coordinated to the

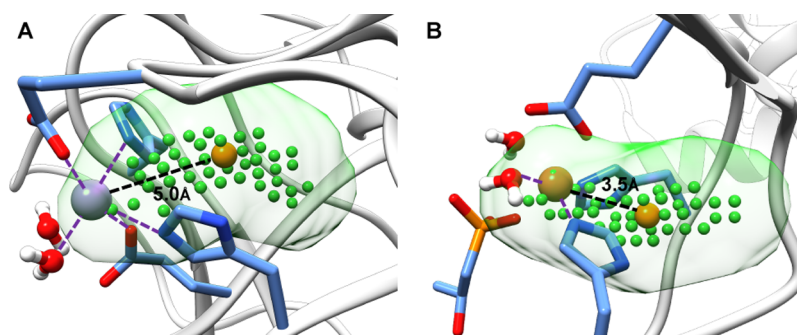


Figure 4. Detail of the binding site for two cases presenting exogenous ligands besides the standard amino acids. In both examples, the center of the predicted area is represented by an orange ball, the probes that form the binding area are represented by smaller green balls surrounded by surface visualization, and the coordination bonds are shown in dashed purple lines. (A) Binding of Mn in PDB 2hxg. The HIS-HIS-GLU-GLU motif is depicted in blue sticks, and two water molecules are highlighted in ball and sticks. (B) Binding of Fe in PDB 1zz7. The HIS-HIS-GLU motif and an exogenous ligand are depicted in blue sticks. Two water molecules are highlighted in ball and sticks.

metal provided by exogenous compounds (e.g., phosphate or water molecules). In these cases, we generally observe a displacement to the limits of the predicted region (Figure 4), suggesting that the exogenous donors only slightly modify the position of the metal and the organization of the side chains. Regarding the computational cost of the calculations, they range from 3 to 245 s, with a median of 18 s (with the size of the system highly correlating with the computational time $r = 0.89$, $p < 0.0001$).

BioMetAll results are therefore particularly promising. It is important to highlight that BioMetAll not only predicts the X-ray structure metal modes but also predicts some that could be different from the experimental structures (from missing one of the amino acids in the coordination sphere to being far from the crystallographic site). Obviously, similar to other techniques based on empirical scores, the predictiveness of the method cannot be limited to the best solution only and those secondary metal-binding sites that BioMetAll detects are not necessarily false positives but transient metallic environments accessible to the protein [see our first application case on human serum albumin (HSA)]. Energetic terms and other physicochemical magnitudes that are not present in BioMetAll could dictate the fine structure of the metal-binding site. Posterior filtering could be necessary depending on the objective of its use.

The benchmark shows that BioMetAll is a suitable piece of software to predict metal-binding sites. One can readily perform screening of protein structures in a rational enzyme design protocol or testing a set of conformations extracted from a molecular dynamics (MD) simulation. With such a tool in hand, a large panel of applications is accessible.

Comparative Study with Other Structure-Based Programs. Although the aim of this article is not an exhaustive review of other software, we found it interesting to compare BioMetAll with other structure-based predictors.^{11–17} We first made a qualitative comparison, evaluating aspects such as code availability or execution time (see Table S6 for details). From this first study, two programs (IonCom and MIB) were selected for further comparative studies because of their recentness, practicability, and better apparent predictivity on a first round of tests. With them, all the calculations of the previous benchmark and the case studies were carried out. Of the 53 binding sites of the FTM benchmark, IonCom was able to detect 37 correctly, while MIB detected 44. Regarding the search of coordinating environments (the exact amino acids that coordinate the metal in the X-ray binding sites), IonCom identified the complete triad of coordinating amino acids in 34 cases, while MIB found the exact crystallographic motif in 30 cases. In 11 cases for IonCom and 2 cases for MIB, the calculation could not be performed mainly because of unavailability of the ion in that program (see Table S7 for the details of the comparative study).

Validation of BioMetAll on a Test Set of New Structures Not Present in the Statistical Analysis. As the structures for the FTM benchmark are included in the data set used for the calculation of geometric constraints, we considered convenient to conduct an additional benchmark on new structures. A search for the crystallographic motif was performed in a set of 40 metal-binding sites where the metal is bound to at least 3 amino acids. The set was obtained by filtering those nonredundant structures with a higher resolution (<1.5 Å) published in the PDB later than when

the data set of MetalPDB was created (September 18th, 2019). Again, all crystallographic motifs were found by BioMetAll, with the other metrics also being comparable to those previously reported for the FTM benchmark (details are provided in Table S8).

Case Study (1): Human Serum Albumin. HSA is the most abundant protein in the blood plasma and, together with human serum transferrin, is the first carrier for most of the essential metal ions.⁴⁵ Two main metal-binding sites have been characterized in HSA: the NTS or ATCUN motif located at the N-terminus⁴⁶ and the multi-metal-binding site (MBS) at the interface of subdomains IA and IIA.⁴⁷ While NTS (with the coordination set $\text{NH}_{2,\text{Asp1}};\text{N}_{\text{Asp1}};\text{N}_{\text{Ala2}};\text{N}_{\text{His3}}$) is preferential for species able to deprotonate amidic hydrogens such as Cu^{2+} and Ni^{2+} , MBS (with the binding $\text{N}_{\text{His67}};\text{N}_{\text{His247}};\text{O}_{\text{Asn99}};\text{COO}_{\text{Asp249}}^{-}$) is the primary site for Zn^{2+} and other bivalent cations.^{48,49} One interesting feature of HSA emerges when comparing its structure in the presence of medium- and long-chain fatty acids (FAs) with those of the defatted state. The binding of FAs to site FA2, which is one of the strongest FA sites located at the same region of MBS, causes the mutual rotation of subdomains IA and IIA, separating the Zn^{2+} -coordinating amino acids and moving a part of Zn^{2+} to NTS with the possible coordination $\text{N}_{\text{His3}}/\text{His9};\text{COO}_{\text{Asp13}}/\text{Glu6}^{-}$.⁵⁰ This conformational change hinders the metal coordination at MBS and has been pointed out in the literature as a regulatory mechanism of zinc homeostasis (Figure 5).^{45,48,51–55}

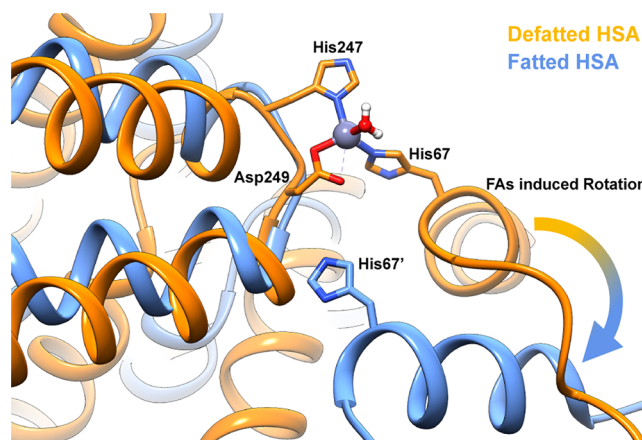


Figure 5. FA-induced rotation, highlighted by the superposition of the IA/IIA interface of HSA in HSA^f (in blue) and HSA^d (in orange) X-ray structures.

Predicting metal-binding sites in front of conformational changes is an interesting challenge for structure-based predictors. Indeed, sequence-based ones would not consider such modification of the protein structure. To ascertain the discriminative capabilities of BioMetAll, both fatted (HSA^f) and defatted (HSA^d) X-ray structures of HSA (PDB codes 1bj5 and 5ijf, respectively) were probed for zones containing a minimum of three coordinator amino acids, whose types should be a combination of His, Asp, and Glu. These are amino acids whose side chains are those with the highest affinity toward Zn^{2+} . A second prediction was performed with a more specific motif similar to the MBS site: two histidine residues and either an aspartate or a glutamate.

As a result of the first analysis, both known primary binding sites (MBS and NTS) were detected by BioMetAll for HSA^d with His-His-Asp/Glu along with other secondary sites that satisfy patterns such as $N_{\text{His}}/\text{COO}_{\text{Asp/Glu}}/\text{COO}_{\text{Glu}}/\text{COO}_{\text{Asp/Glu}}^{-}$ (Table S9) and that have also been reported in the literature.⁵⁰ However, for HSA^f, only NTS (His3-His9-Glu6) together with the same secondary sites than HSA^d is identified (Table S10). By looking for the specific motif His-His-Asp/Glu, MBS and NTS were uniquely identified for HSA^d (Figure 6A), while

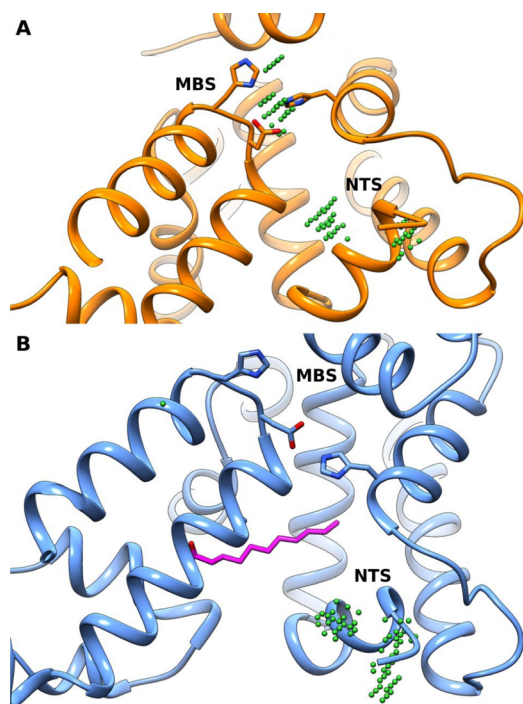


Figure 6. Probes (green dots) predicted by BioMetAll by looking for the motif His-His-Asp/Glu. (A) Defatted conformation, in which the experimental binding site is predicted with high precision. (B) Fatted conformation, where the experimental binding site is no longer predicted. FA bound to FA2 is depicted in purple.

only the NTS binding site was detected for HSA^f (Figure 6B). Hence, BioMetAll differentiates the metal-binding profile between both conformational states of HSA. Besides, BioMetAll is sensitive enough to still detect the incomplete MBS in HSA^f (His247, Asp249) if a reduced number of

coordinators is used. When it comes to the comparison with other programs, IonCom could not detect MBS in either of the two conformations, while MIB detected the same two coordinating residues (His247, Asp249) for both HSA^f and HSA^d.

This case study points out that BioMetAll is not only able to predict with accuracy primary and secondary metal-binding sites but also able to account for the impact of conformational changes of the protein and how it affects its preorganization for metal binding.

Case Study (2): Hemocyanins. Our second illustrative case focuses on copper-binding proteins. From hemocyanin to blue copper protein, nature has intensively adapted its toolbox to incorporate this metal. Interestingly, copper is a very selective transition metal in terms of amino acids to which it could bind. Statistics (Table S3) show that copper highly prefers histidine (70.8%). Other amino acids such as cysteines (16.3%) or methionines (5.6%) are found with far lower frequencies. Glutamates (3.5%), as well as aspartates (2.3%), end the list of main coordinating amino acids to copper. One could therefore hypothesize that a way to detect copper-binding sites consists in identifying those histidine residues that are well preorganized for metal binding.

We performed BioMetAll runs on a series of hemocyanin proteins – a prototypical family of di-copper-binding oxygen carriers – that coordinate copper throughout motifs of three histidine amino acids. After removal of the metal ions from the .pdb files, a set of experimentally characterized structures were screened to look for sites with a minimum of three histidine residues able to coordinate a metal ion. The search led to a 100% prediction for the di-copper-binding sites for all the hemocyanins analyzed at the exact location of the X-ray structure (Table S11). One example is given in Figure 7 with *Manduca sexta* prophenoloxidase, a heterodimer consisting of two homologous polypeptide chains.⁵⁶

One of the advantages of BioMetAll is to provide the user to control the specificities of the metal-binding search pattern. Even those that are not the “best-case scenario” can be evaluated based, for example, on chemical hypothesis. With such an idea in mind, we performed a second series of calculations releasing the restraint of the search motif from three histidine residues to only two with eventually a third amino acid being cysteine, methionine, aspartate, or glutamate. The results provide with several putative metal-binding sites spreading to a broader part of the protein, but, to our surprise, these were not randomly scattered inside the scaffold (an

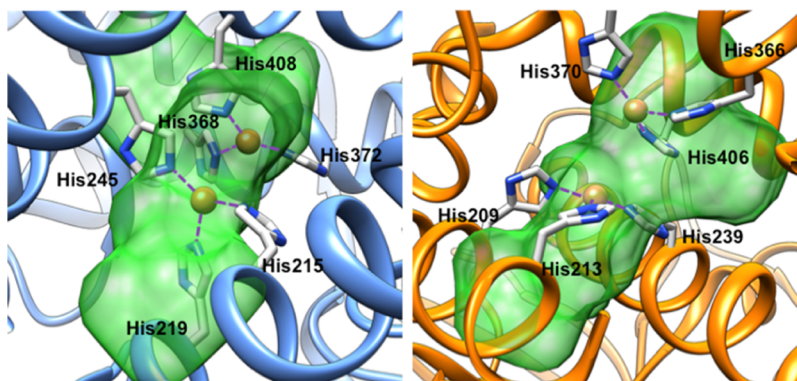


Figure 7. Di-copper-binding sites for the two monomers (in blue and orange ribbons) of *M. sexta* prophenoloxidase (PDB code 3hhs). Probes predicted by BioMetAll are shown in green surfaces and amino acids participating in copper binding are highlighted in white sticks.

example for *M. sexta* prophenoloxidase is shown in Figure 8). Instead, a continuum pattern of adjacent sites that evolves

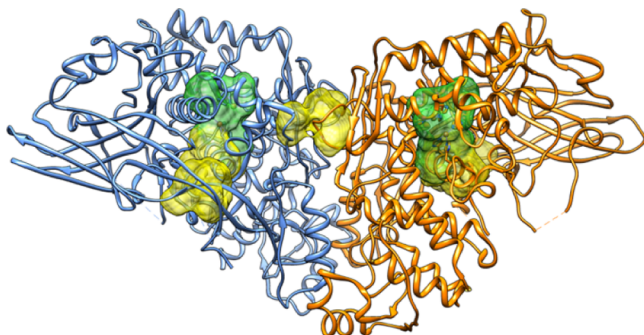


Figure 8. Overall view of the structure of *M. sexta* prophenoloxidase, with the results of BioMetAll calculations overlapped. Probes for the three-histidine calculation are shown in green surfaces, and probes for the two-histidine calculation are shown in yellow surfaces. Continuous paths from the solvent to the di-copper-binding sites are observed.

from solvent-exposed locations to the buried dicopper site was observed. This observation, combined with the presence of water molecules along those channels in several crystal structures, strongly suggests the possibility that BioMetAll allowed the detection of metal diffusion pathways.

It is interesting to comment that other structure-based predictors (i.e., IonCom and MIB) were able to identify the crystallographic binding site but did not obtain the transient sites that make up the access channel. Also, our intents to use traditional approaches for channeling predictions, such as those based on calculating solvent-excluded surface calculations, did not lead to such clear prediction of metal-compatible pathways, nor a map of histidine onto the crystallographic structure would provide with such information (although being obviously related). Although falling far for the main objectives of this work, such observation could be particularly interesting to investigate the properties of metal diffusion in hemocyanins and their relationship to oligomeric states. The versatility of BioMetAll could lead to interesting outcomes when looking for diffusion pathways of metallic species in proteins.

Case Study (3): Metalloenzyme Design. Because BioMetAll is mostly based on backbone patterns, prediction of mutations to generate metal-binding sites is probably the most interesting application. For our last illustrative case, we aimed at showing the interest of this approach for metalloenzyme design. To do so, the repurposing study of a hydrolase to a peroxidase by Ward et al.⁵⁷ was revisited.

The original work by Ward and co-workers started with a screening of the PDB³⁸ for FTM-compatible motifs including possible mutations from Asn/Gln to Asp/Glu. With an initial screening of the entire PDB with the software STAMPS,⁵⁸ six structures were identified as interesting candidates and followingly tested for peroxidase activity. Only 6-phosphogluconolactonase from mycobacterium smegmatis (whose structure is accessible at the PDB with the code 3oc6) with an Asn131Asp mutation showed catalysis in the presence of CuSO₄. Interestingly, X-ray crystallography of the resulting system enlightened that only the His67-His104 tandem with histidine *trans* to each other coordinates the copper ion; the predicted de novo aspartate at the 131 position was not able to coordinate.

To show the interest of BioMetAll in an artificial metalloenzyme framework, we started by looking for the His-His-Asp/Glu motif in the 6-PGLac structure. In agreement with the initial study, no positive outcomes came out. The search was therefore expanded to His-His-Asn/Gln motifs obtaining the His67-His104-Asn131 triad found by Ward et al. The subsequent analysis was searching for any mutation that could complete the His67-His104 pair with a Glu/Asp. Unlike other structural prediction programs that need the exact sequence to be present in the target protein, BioMetAll directly outputs a list of mutations. Thus, it was not possible to compare with IonCom or MIB in this case.

This run of BioMetAll looking for possible sites allowing a mutation to either Asp or Glu provided with an extensive list of possible solutions including Asn131 (Table 5). The following

Table 5. Summary of Mutations Proposed by BioMetAll to Achieve a His-His-Glu/Asp in the 6-PGLac Structure, Starting from the Already Present His67-His104^a

number	mutated a.a.	Asp ^b	Glu ^c
1	Tyr69	18 (6)	34 (16)
2	Thr38	11 (5)	31 (13)
3	Asn131	8 (0)	29 (9)
4	Ile68 ^d	3 (1)	18 (8)
5	Ile39 ^d	2 (0)	9 (5)
6	Val101	2 (2)	18 (10)
7	Val40 ^d	2 (1)	10 (3)
8	Asn102	1 (1)	19 (14)
9	Leu127 ^d	0	7 (3)
10	Leu128	0	2 (0)
11	Phe139	0	2 (0)
12	Val103	0	2 (0)
13	Leu41	0	1 (0)
14	Pro99	0	1 (0)

^aIn bold are the possibilities taken into account in the docking assay.

^bNumber of probes for the binding area when mutating the a.a. to Asp is reported; in parenthesis is the number of probes after filtering.

^cNumber of probes for the binding area when mutating the a.a. to Glu is reported; in parenthesis is the number of probes after filtering.

^dSolutions excluded because of visual inspection (see Figure 9A).

step was an additional execution of the program to filter those probes that show collapse with any amino acid side chain or backbone from close neighbors of the putative metal-binding areas. This filtering process together with a visual inspection diminished the number of possibilities to five candidates (Table 5 and Figure 9A).

To further assess the predictiveness of BioMetAll under a metalloenzyme design framework, the predicted systems were used as a starting point for docking experiments. Calculations were carried out with the protein–ligand docking software GOLD^{59–61} and repeated inside our GaudiMM platform (to provide with a freeware alternative).^{32,62} Both calculations were performed following our protocols optimized for metalloligands binding to proteins^{30,31} (see the Supporting Information for further details). The list of FTMs tested includes therefore His67-His104-Asp/Glu69, His67-His104-Asp/Glu38, His67-His104-Glu131, His67-His104-Asp/Glu101, and His67-His104-Asp/Glu102. For the sake of the comparison with the initial work and the consequent X-ray structure, His67-His104-Asp131 was also analyzed. From all the calculations performed, solutions presenting convenient

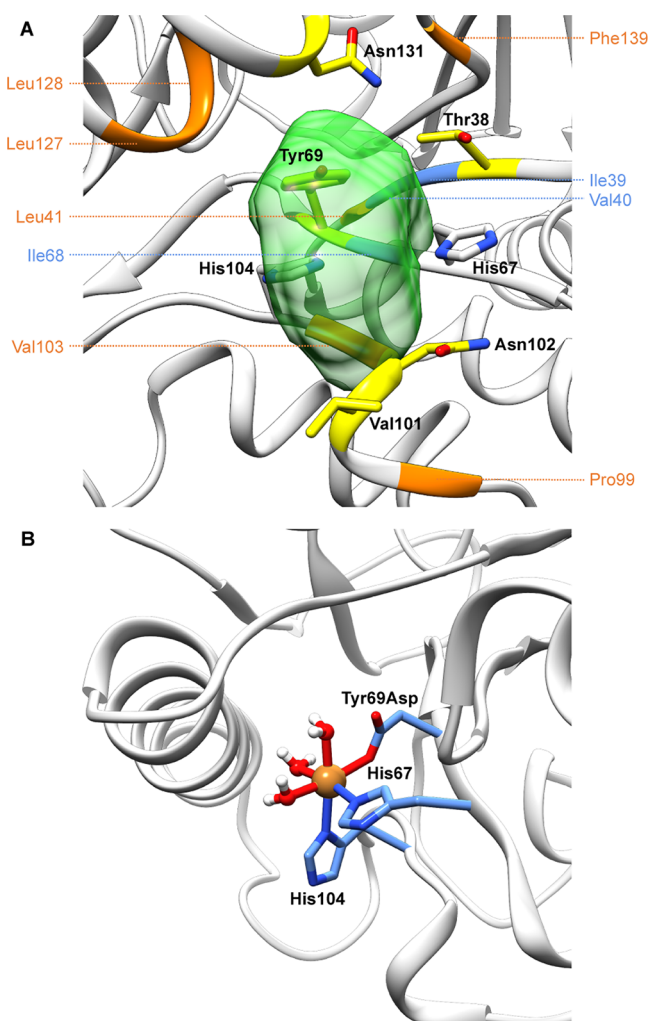


Figure 9. (A) Set of feasible mutations predicted by BioMetAll to add an Asp/Glu to the His67-His104 motif. The five options considered in further docking analysis (Tyr69, Thr38, Asn131, Val101, and Asn102) are depicted in yellow sticks. The discarded mutations are highlighted in orange (options for Glu) or blue (options for Asp/Glu) ribbons. The probe area predicted as feasible for metal binding is shown in green surface. (B) Best docking pose obtained for Tyr69Asp mutation, where the copper is facially coordinated by the His67-His104 triad.

metal-binding modes were only observed for the mutation Tyr69Asp/Glu, even though the Asp mutation reaches a better coordination geometry in the docking assay. In this case, the FTM motif appears excellently reproduced with the side chains displaying geometries that allow the aspartate to bind the metal facially (Figure 9B). It must be mentioned that mutations could affect the general folding and functions of the protein. This aspect should be evaluated either by a subsequent MD method or by other methods such as prediction of deleterious synonymous mutations.⁶³ Finally, we want to stress that only those mutations with about 10 probes and more appear as viable for coordination, with the mutation at position 69 being outstanding for both Asp and Glu mutations. This finding suggests that the number of probes could be used as a criterion to prioritize possible proposed solutions. Strikingly, no docking calculation allows producing a convenient binding mode of the FTM motif for the Asn131Asp mutation with no rearrangement of the His67-His104-Asp131 side chains to coordinate

with the metal. This result sustains the BioMetAll prediction regarding the low potential of such a mutant to conveniently coordinate the metal as well as with the experimental X-ray structure.

From this part of the study, BioMetAll and its combination with docking experiments optimized for metal binding appear to predict convenient side-chain rearrangements. For the purpose of this, the enzyme design exercise should only be viable for the Tyr69Asp and to a less extent Tyr69Glu.

CONCLUSIONS

The large number of experimental structures available in the PDB offers today a unique landscape to understand the molecular rules that govern the binding of metallic moieties to proteins and provide grounds to generate structural predictors. In this work, we show that it is possible to build an efficient predictor of metal-binding sites based on a few geometric descriptors of the conformation of the backbone. Our intent, BioMetAll, gives a major weight of the prediction to the preorganization capacities of the protein.

BioMetAll shows great potential, supported by tests on several systems, showing that (i) it efficiently predicts primary as well as secondary metal-binding sites, (ii) it can easily account for how the conformational changes of the protein could alter the formation of metal-binding sites, (iii) it allows to work with common metal-binding templates as well as nonstandard ones, (iv) it identifies sites even with incomplete coordination spheres, and (v) it allows to predict mutations that could be necessary to generate new metal-binding sites in a protein, for example, for building new biocatalysts. BioMetAll predictions are also of good quality when exogenous compounds are also involved in the experimental binding site.

BioMetAll's novel philosophy differs substantially from other structure-based predictors that generally give a major weight to detect perfect matching between the target and known geometries of the first coordination sphere of the metal. From the comparison with other tools in the systems analyzed in this work, it is revealed that BioMetAll's performance compares very well when it comes to determining metal-binding sites as well as the exact list of the amino acids involved in the coordination motifs.

BioMetAll, and its ground on backbone preorganization, appears as a very interesting option in fields related to biometallic interactions. It is fundamental to highlight that BioMetAll does not consider energetic terms. Posterior refinements might be necessary depending on the object of the study. At the moment, we put a large amount of effort to bridge BioMetAll to other methods such as those allowing extensive conformational sampling such as MD and protein-ligand dockings. We believe that this kind of combination would open new horizons for research in chemistry and its interface with biology.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.0c00827>.

Computational details; number of coordination bonds obtained after processing the MetalPDB database; ranges for each one of the geometric features analyzed, grouped by the type of amino acid; number of coordination bonds involving a side-chain atom as a

donor; number of coordination bonds involving a backbone oxygen atom as a donor; results obtained with BioMetAll for the benchmark searching for HIS-HIS-ASP/GLU motifs; qualitative comparison with other structure-based metal-binding site predictors; results obtained with IonCom and MIB for the benchmark searching for FTMs; results obtained with BioMetAll for the validation benchmark searching for experimental motifs on new structures; BioMetAll results for the search of a minimum of three coordinator amino acids (combination of His, Asp, and Glu) on HSA^d; BioMetAll results for the search of a minimum of three coordinator amino acids (combination of His, Asp, and Glu) on HSA^f; and summary of the copper-binding centers in different hemocyanins (PDF)

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Notes

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